

Oluwatomisin Adenekan

San Mateo, CA | (650)-293-1384 | tominekan12@gmail.com | <https://tominekan.netlify.app> | LinkedIn

EDUCATION

University Of Pennsylvania – Cumulative GPA: 3.62

Philadelphia, PA

Bachelor of Science in Engineering

Expected Graduation: May 2028

Majors: Computer Science, Philosophy

Relevant Coursework: Applied Machine Learning, Computer Vision, Big Data Analytics, Databases, Operating Systems

TECHNICAL SKILLS

Python, Pandas, C, Java, HTML/CSS/JS, TypeScript, JavaScript, Git/GitHub

EXPERIENCE

Microsoft

Philadelphia, PA

Software Engineering Intern

May 2026 – Aug 2026

- Architected a full-stack AI agent (React, C# .NET) within self-hosted government clouds, streamlining financial data retrieval by replacing 17 rigid legacy scripts with a unified natural language interface
- Modernized deployment reporting workflows, allowing stakeholders to bypass hardcoded queries and instantly extract complex, ad-hoc financial insights
- Developed a full-stack employee table tennis match scheduling application (React, Python/FastAPI, SQLAlchemy, Azure) with an ELO ranking system, intelligent match recommendations, live leaderboards, and optimized resource allocation

Weingarten Center

Philadelphia, PA

Peer Tutor

Aug 2025 – Present

- Selected to tutor Math 1400 (Single Variable Calculus), CIS 1200 (Programming Languages and Techniques) and Math 2400 (Linear Algebra) based on strong academic performance in the course
- Provide one-on-one and group tutoring for 30+ students, assisting with difficult concepts, problem-solving, and exam preparation

Center for Neuroengineering & Therapeutics

Philadelphia, PA

Research Assistant

May 2025 – Present

- Developed a logistic regression-based web tool to predict seizure laterality, aimed at improving surgical planning
- Applied signal processing and statistical analysis using Python to EEG data, identifying seizure onset zones with 80% accuracy
- Engineered scalable data pipelines (Bash, Batch, Python) to automate processing of 226GB+ of EEG recordings
- Use Pandas to analyze spike rates within interictal EEG data to determine seizure-onset patterns
- Co-authored “Quantifying Interictal Spike Asymmetry on Scalp EEG to Lateralize the Seizure-Onset Zone in Temporal Lobe Epilepsy”, a publication in *Neurology Open Access*

Fife-Penn Stem and CS Academy

Philadelphia, PA

Instructor

Sep 2024 – Present

- Design and implement weekly curriculum for 75+ elementary school children in various high schools across Philadelphia
- Teach Scratch through custom-made workshops and demonstrative games, improving student engagement with STEM concepts

LEADERSHIP & PROFESSIONAL DEVELOPMENT

Penn Sports Analytics Group

Philadelphia, PA

Club Member

Aug 2025 – Present

- Collaborate with Penn’s Strength and Conditioning coaches to rank, evaluate, and predict player performance across several sports using research-based metrics like the Total Score of Athleticism (TSA)

SEO Career

Remote, USA

SEO Edge Participant

Feb 2024 – Present

- Receive coaching and online instruction to achieve targeted interview, technical, and professional developmental goals

PROJECTS

BabyDB

June 2026

- Build a columnar database in C using mmap and SIMD, achieving a 54-microsecond load latency
- Design a distributed ZeroMQ replication layer, optimizing CPU cache locality via strict 64-row memory alignment

Image Style Transfer

May 2026

- Developed a geometry-preserving style transfer pipeline, fine-tuning a Canny edge-conditioned Stable Diffusion model via LoRA to translate weather and lighting while strictly maintaining structural consistency
- Built an automated dataset curation pipeline using LightGlue for precise feature matching, successfully aligning, deduplicating, and standardizing 240+ raw images into a curated Hugging Face dataset

PyComplete

Aug 2025

- Developed a text autocomplete library and API using Python and FastAPI, leveraging Markov chain algorithms
- Designed and implemented a mock E-Commerce frontend in React to validate the PyComplete API’s performance